

optimization could plausibly reduce these degradations. Overall, these results are comparable to quality losses reported for traditional watermarking approaches in previous work (Yang et al., 2025).

G. Cross-Dataset and Cross-Model Generalization

To assess out-of-distribution robustness, we ran cross-evaluation experiments along two axes. The training pipeline is defined as:

$$\text{train: } \text{Prompt}_1 \in D_1 \rightarrow \text{LLM}_a \rightarrow \text{Text}_{a1} \rightarrow \text{LLM}_a \rightarrow A_{a1} \rightarrow \text{MLP}_a$$

Domain adaptation tests use a different dataset with the same model:

$$\text{test: } \text{Prompt}_2 \in D_2 \rightarrow \text{LLM}_a \rightarrow \text{Text}_{a2} \rightarrow \text{LLM}_a \rightarrow A_{a2} \rightarrow \text{MLP}_a$$

Model adaptation tests use a different generating LLM, with activations extracted from the original training model:

$$\text{test: } \text{Prompt}_1 \in D_1 \rightarrow \text{LLM}_b \rightarrow \text{Text}_{b1} \rightarrow \text{LLM}_a \rightarrow A_{b \rightarrow a} \rightarrow \text{MLP}_a$$

In both cases, activation collection and MLP inference always use the training setup, reflecting a realistic deployment scenario where the provider controls the detection pipeline. We report in Table 9 token-level F1 and aggregated text-level F1 scores for binary classification between two distinct steering vectors.

Table 9. Cross-evaluation F1 scores (token-level / text-level) for binary steering vector classification across training and test configurations. Bold diagonal entries indicate in-distribution evaluation.

Train Setup		Test Setup					
		Llama-3.1-8B-Instruct		Llama-3.1-8B		Ministral-3-8B-Base	
		ELI5	News	News	ELI5	News	ELI5
Llama-3.1-8B-Instruct	ELI5	0.90/0.99	0.77/0.84	0.74/0.80	0.74/0.85	0.55/0.52	0.58/0.51
Llama-3.1-8B	News	0.76/0.84	0.86/0.94	0.90/0.99	0.78/0.90	0.56/0.61	0.61/0.65

As expected, out-of-distribution performance degrades, though models trained on higher-entropy data generalize better across both domain and architecture shifts, likely due to richer activation variability during training. Notably, the steering signal partially survives even across the Llama-3.1-8B Base and Instruct versions ($\text{LLM}_b \rightarrow \text{LLM}_a$), suggesting that steering leaves traces beyond a single checkpoint within the same architecture. This no longer holds when generating data with a different architecture, such as a Ministral-3-8B model. This matches the results obtained in subliminal learning (Cloud et al., 2025), where subliminal information is transmitted only within similar architectures. Training on a diverse mixture of prompt types and model variants is a natural path to a more robust detector.

Cross-model transfer for self-recognition. We similarly evaluate cross-model generalization for the self-recognition classifiers introduced in Section 3.1. For each training model, we extract residual-stream activations from that same model and train a classifier on texts generated by it. At test time, the training model and classifier are kept fixed, but the evaluated texts are generated by a different model. Thus, only the text-generating model changes across columns in Table 10, while activation extraction and classifier inference remain fixed.

We observe that cross-model performance decreases relative to the in-distribution setting, with the strongest degradation appearing when the classifier trained on Ministral-3-8B activations is evaluated on Llama-family generations. In contrast, classifiers trained on Llama-family activations transfer more consistently across Llama-generated texts, suggesting stronger alignment of activation-space signatures within related architectures. These results indicate that self-recognition through residual-stream activations captures model-specific traces that are only partially shared across LLMs. Consequently, cross-model robustness remains limited in this setting, and training on activations induced by a mixture of model families is a natural extension for improving transfer.

Table 10. Cross-model classification accuracy for self-recognition based on residual-stream activations. Rows correspond to the model used for activation extraction and classifier training, while columns correspond to the model that generated the evaluated texts. Italicized diagonal entries indicate in-distribution evaluation.

Train Setup	Test Setup			
	Ministral-3-8B	Llama-3.1-8B	Llama-3.2-3B	Llama-3.2-1B
Ministral-3-8B	<i>0.9986</i>	0.8089	0.8068	0.7709
Llama-3.1-8B	0.9865	<i>0.9800</i>	0.9633	0.9354
Llama-3.2-3B	0.9860	0.9645	<i>0.9798</i>	0.9428
Llama-3.2-1B	0.9807	0.9493	0.9510	<i>0.9772</i>